Grokking Modern System Design Interview

System Design Interviews decide your level and compensation at top tech c

First Edition

PREFACE

This book was automatically generated from the Educative course: **Grokking Modern System Design Interview** .

Course Overview

Everything you need for Grokking the System Design Interview, developed by FAANG engineers. Master distributed system fundamentals \ practice real-world interview questions.

What You'll Learn

System Design Interviews decide your level and compensation at top tech companies. To succeed, you must design scalable systems, justify trade-offs, and explain decisions under time pressure. Most candidates struggle because they lack a repeatable method.

Built by FAANG engineers, this is the definitive System Design Interview course. You will master distributed systems building blocks: databases, caches, load balancers, messaging, microservices, sharding, replication, and consistency, and learn the patterns behind web-scale architectures.

Using the RESHADED framework, you will translate open-ended system design problems into precise requirements, explicit constraints, and success metrics, then design modular, reliable solutions. Full Mock Interview practice builds fluency and timing.

By the end, you will discuss architectures with Staff-level clarity, tackle unseen questions with confidence, and stand out in System Design Interviews at leading companies.

About This Generated Book

This content has been automatically processed and formatted from the original Educative interactive course.

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How to Use This Book

This book is structured to follow the original course progression. Each chapter corresponds to a major section of the course, with individual lessons presented as sections within each chapter.

For the best learning experience, consider accessing the original interactive course on Educative.io, which includes:

- Interactive coding environments
- Hands-on exercises and quizzes
- Real-time feedback and hints
- Practical projects and assessments

Happy Learning!

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Explore what System Design interviews involve, including essential preparation strategies, fundamental concepts, key resources, and tips to perform well.

1 Getting Ready for the System Design Interview

Section content for "Getting Ready for the System Design Interview" is being generated.

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- Section Slug: key-concepts-to-prepare-for-the-system-design-interview
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Get familiar with System Design and the course structure, including prerequisites that set the foundation for mastering System Design interviews.

1 Introduction to Modern System Design

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CHAPTER 3

ABSTRACTIONS

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Grasp the fundamentals of abstractions in distributed systems, focusing on network abstraction, consistency, and failure models crucial for System Design.

1 Why Are Abstractions Important?

1.1 What is abstraction?

Abstraction is the art of obfuscating details that we don't need. It allows us to concentrate on the big picture. Looking at the big picture is vital because it hides the inner complexities, thus giving us a broader understanding of our set goals and staying focused on them. The following illustration is an example of abstraction.

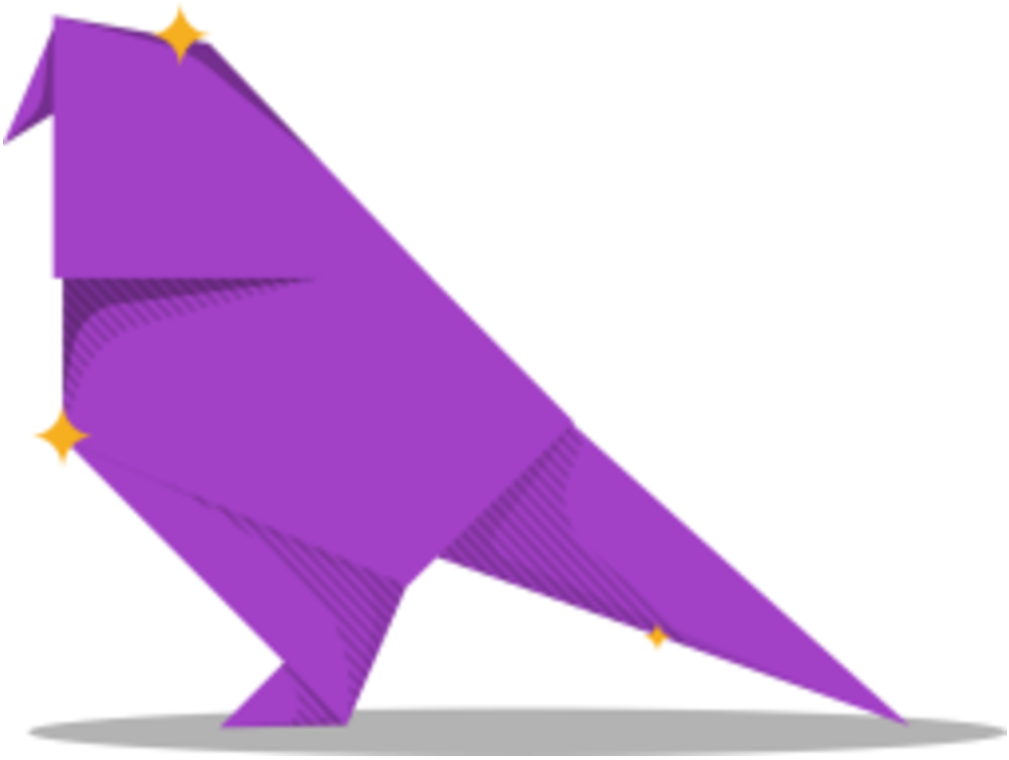


Figure 3.1: Abstraction of a bird

With the abstraction shown above, we can talk about birds in general without being bogged down by the details.

> **Note:** If we had drawn a picture of a specific bird or its features, we wouldn't achieve the goal of recognizing all birds. We'd learn to recognize a particular type of bird only.

In the context of computer science, we all use computers for our work, but we don't start making hardware from scratch and developing an operating system. We use that for the purpose at hand rather than digging into building the system.

The developers use a lot of libraries to develop the big systems. If they start building the libraries, they won't finish their work. Libraries give us an easy interface to use functions and hide the inside detail of how they are implemented. A good abstraction allows us to reuse it in multiple projects with similar needs.

1.2 Database abstraction

Transactions is a database abstraction that hides many problematic outcomes when concurrent users are reading, writing, or mutating the data and gives a simple interface of commit, in case of success, or abort, in case of failure. Either way, the data moves from one consistent state to a new consistent state. The transaction enables end users to not be bogged down by the subtle corner-cases of concurrent data mutation, but rather concentrate on their business logic.

1.3 Abstractions in distributed systems

Abstractions in distributed systems help engineers simplify their work and relieve them of the burden of dealing with the underlying complexity of the distributed systems.

The abstraction of distributed systems has grown in popularity as many big companies like Amazon AWS, Google Cloud, and Microsoft Azure provide distributed services. Every service offers different levels of abstraction. The details behind implementing these distributed services are hidden from the users, thereby allowing the developers to focus on the application rather than going into the depth of the distributed systems that are often very complex.

Today's applications can't remain responsive/functional if they're based on a single node because of an exponentially growing number of users. Abstractions in distributed systems help engineers shift to distributed systems quickly to scale their applications.

> **Note:** We'll see the use of abstractions in communications, data consistency, and failures in this chapter. The purpose is to convey the core ideas, but not necessarily all the subtleties of the concepts.

2 Network Abstractions: Remote Procedure Calls

Remote procedure calls (**RPCs**) provide an abstraction of a local procedure call to the developers by hiding the complexities of packing and sending function arguments to the remote server, receiving the return values, and managing any network retries.

2.1 What is an RPC?

RPC is an interprocess communication protocol that's widely used in distributed systems. In the OSI model of network communication, RPC spans the transport and application layers.

RPC mechanisms are employed when a computer program causes a procedure or subroutine to execute in a separate address space.

Note: The procedure or subroutine is coded as a regular/local procedure call without the programmer explicitly coding the details for the remote interaction.

2.2 How does RPC work?

When we make a remote procedure call, the calling environment is paused and the procedure parameters are sent over the network to the environment where the procedure is to be executed.

When the procedure execution finishes, the results are returned to the calling environment where execution restarts as a regular procedure call.

To see how it works, let's take an example of a client-server program. There are five main components involved in the RPC program, as shown in the following illustration:

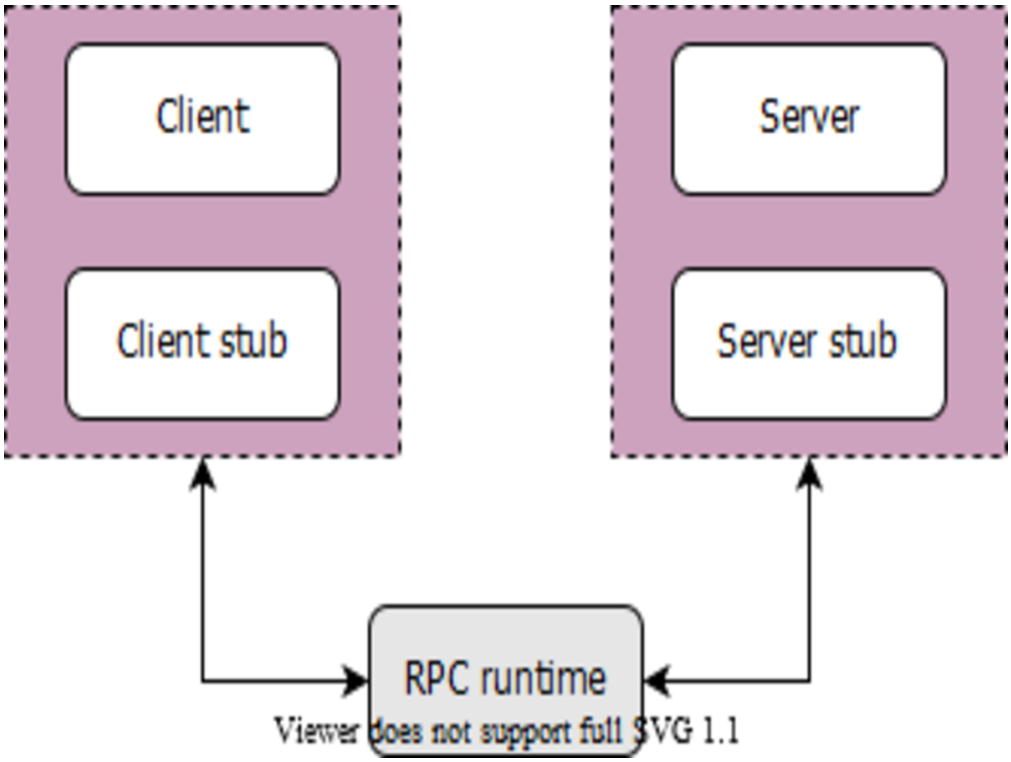


Figure 3.2: The components of an RPC system

The client, the client stub, and one instance of RPC runtime are running on the client machine. The server, the server stub, and one instance of RPC runtime are running on the server machine.

During the RPC process, the following steps occur:

1. A client initiates a client stub process by giving parameters as normal. The client stub is stored in the address space of the client.

2. The client stub converts the parameters into a standardized format and packs them into a message. After packing the parameter into a message, the client stub requests the local RPC runtime to deliver the message to the server.

3. The RPC runtime at the client delivers the message to the server over the network. After sending a message to the server, it waits for the message result from the server.

4. RPC runtime at the server receives the message and passes it to the server stub.

   Note: The RPC runtime is responsible for transmitting messages between client and server via the network. The responsibilities of RPC runtime also include retransmission, acknowledgment, and encryption.

5. The server stub unpacks the message, takes the parameters out of it, and calls the desired server routine, using a local procedure call, to do the required execution.

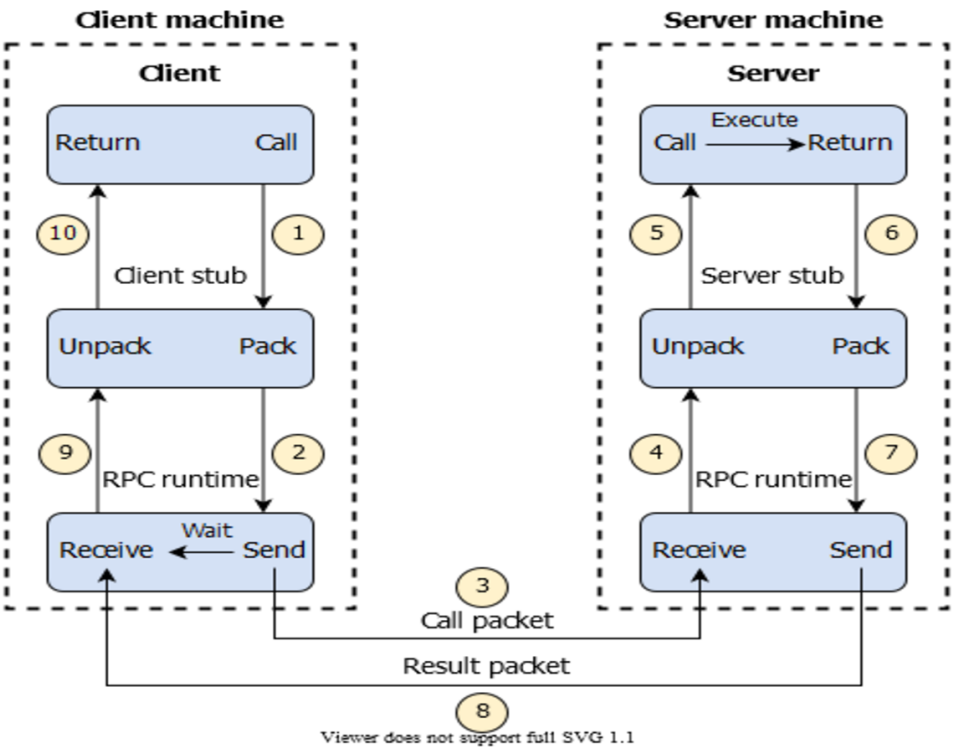


Figure 3.3: The workflow of an RPC

6. After the server routine has been executed with the given parameters, the result is returned to the server stub.

7. The server stub packs the returned result into a message and sends it to the RPC runtime at the server on the transport layer.

8. The server's RPC runtime returns the packed result to the client's RPC runtime over the network.

9. The client's RPC runtime that was waiting for the result now receives the result and sends it to the client stub.

10. The client stub unpacks the result, and the execution process returns to the caller at this point.

    Note: Back-end services use RPC as a communication mechanism of choice due to its high performance and simple abstraction of calling remote code as local functions.

Real-world usage

RPCs are used in many real-world services. Take a look at the examples given below:

- **Google:** Google uses RPC in various parts of its distributed infrastructure. They have developed gRPC, an open-source framework that uses RPC to build efficient and highly performant distributed systems. It's employed for services like Google Search, YouTube, etc., where it allows Google to maintain communication and data exchange between different components of their services, which is essential for achieving real-time performance.

- **Uber:** Uber uses RPC for various functions, including real-time location tracking, ride matching, and communication between drivers and riders. RPC allows Uber to provide fast and responsive services by facilitating data exchange between the user's app and the server.

- **Facebook:** Most services at Facebook are written using Thrift for RPC, and some storage systems use Thrift for serializing records on disk. This provides several benefits to Facebook. For example, it enables interoperability between different languages, such as a Python client communicating with a C++ server.

2.3 Summary

The RPC method is similar to calling a local procedure, except that the called procedure is usually executed in a different process and on a different computer.

RPC allows developers to build applications on top of distributed systems. Developers can use the RPC method without knowing the network communication details. As a result, they can concentrate on the design aspects, rather than the machine and communication-level specifics.

3 Spectrum of Consistency Models

3.1 What is consistency?

In distributed systems, consistency may mean many things. One is that each replica node has the same view of data at a given point in time. The other is that each read request gets the value of the recent write. These are not the only definitions of consistency, since there are many forms of consistency. Normally, **consistency models** provide us with abstractions to reason about the correctness of a distributed system doing concurrent data reads, writes, and mutations.

If we have to design or build an application in which we need a third-party storage system like S3 or Cassandra, we can look into the consistency guarantees provided by S3 to decide whether to use it or not. Let 's explore different types of consistency.

The two ends of the consistency spectrum are:

- *Strongest consistency*
- *Weakest consistency*

There are consistency models that lie between these two ends, some of which are shown in the following illustration:

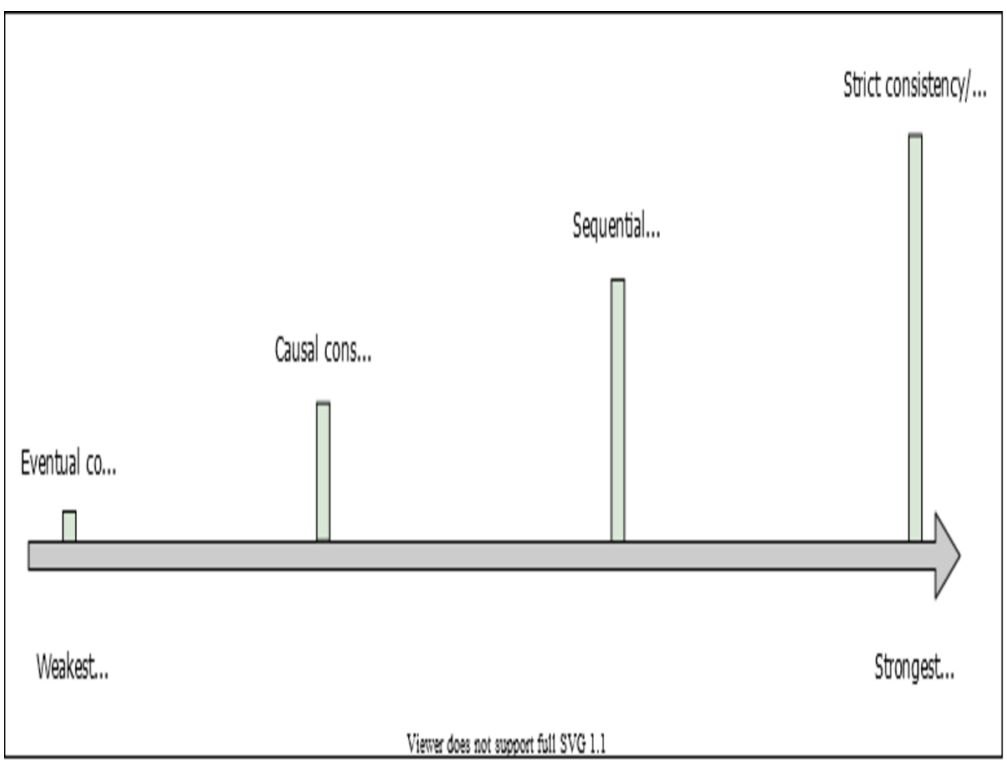


Figure 3.4: This is a spectrum of consistency models. Consistency guarantees get stronger as we move to the right

There is a difference between consistency in *ACID* Atomicity, Consistency, Isolation, and Durability *properties* and consistency in the *CAP theorem*.

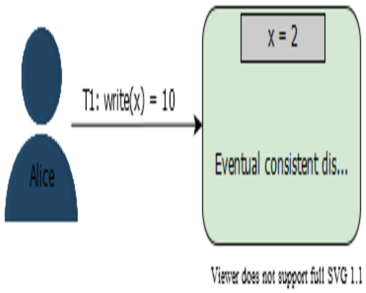
Database rules are at the heart of **ACID consistency**. If a schema specifies that a value must be unique, a consistent system will ensure that the value is unique throughout all actions. If a foreign key indicates that deleting one row will also delete associated rows, a consistent system ensures that the state can't contain related rows once the base row has been destroyed.

CAP consistency guarantees that, in a distributed system, every replica of the same logical value has the same precise value at all times. It's worth noting that this is a logical rather than a physical guarantee. Due to the speed of light, replicating numbers throughout a cluster may take some time. By preventing clients from accessing different values at separate nodes, the cluster can nevertheless give a logical picture.

3.2 Eventual consistency

Eventual consistency is the weakest consistency model. The applications that don't have strict ordering requirements and don't require reads to return the latest write choose this model. Eventual consistency ensures that all the replicas converge on a final value after a finite time and when no more writes are coming in. If new writes keep coming, replicas of an eventually consistent system might never reach the same state. Until the replicas converge, different replicas can return different values.

(a) Initially, the value of x is 2



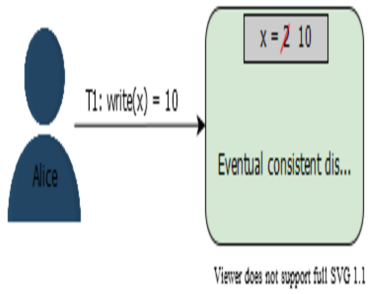
(b) At time T1, Alice sends a write request to update the value of x to 10

Eventual consistency ensures **high availability**.

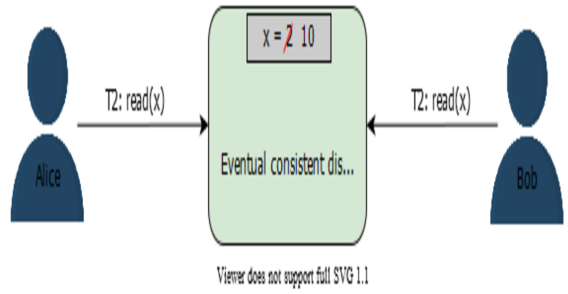
Example

The **domain name system** is a highly available system that enables name lookups to a hundred million devices across the Internet. It uses an eventual consistency model and doesn't necessarily reflect the latest values.

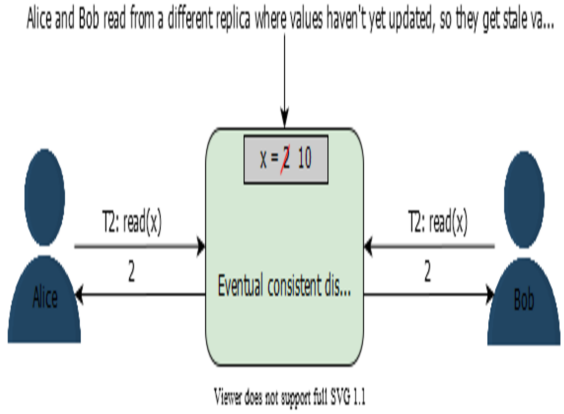
> **Note:** Cassandra is a highly available NoSQL database that provides eventual consistency.



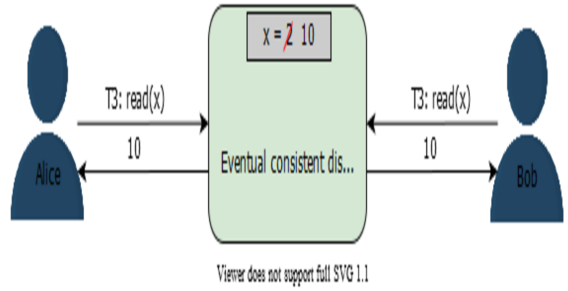
(a) The system saves the updated value of x



(b) At time T2, Alice and Bob read the value of x



(a) The system returns the same value to both read requests, but the value is old



(b) At time T3, system reaches the stable state and returns the latest value to both read requests

3.3 Causal consistency

Causal consistency works by categorizing operations into dependent and independent operations. **Dependent operations** are also called causally-related operations. Causal consistency preserves the order of the causally-related operations.

In the following illustration, process P1 writes a value a at location x. For P2 to write the value b at location y, it first needs to calculate b. Since $b=x+5$, the read operation on x should be performed before writing b on location y. That 's why *read(x)a* and *write(y)b* are causally related.

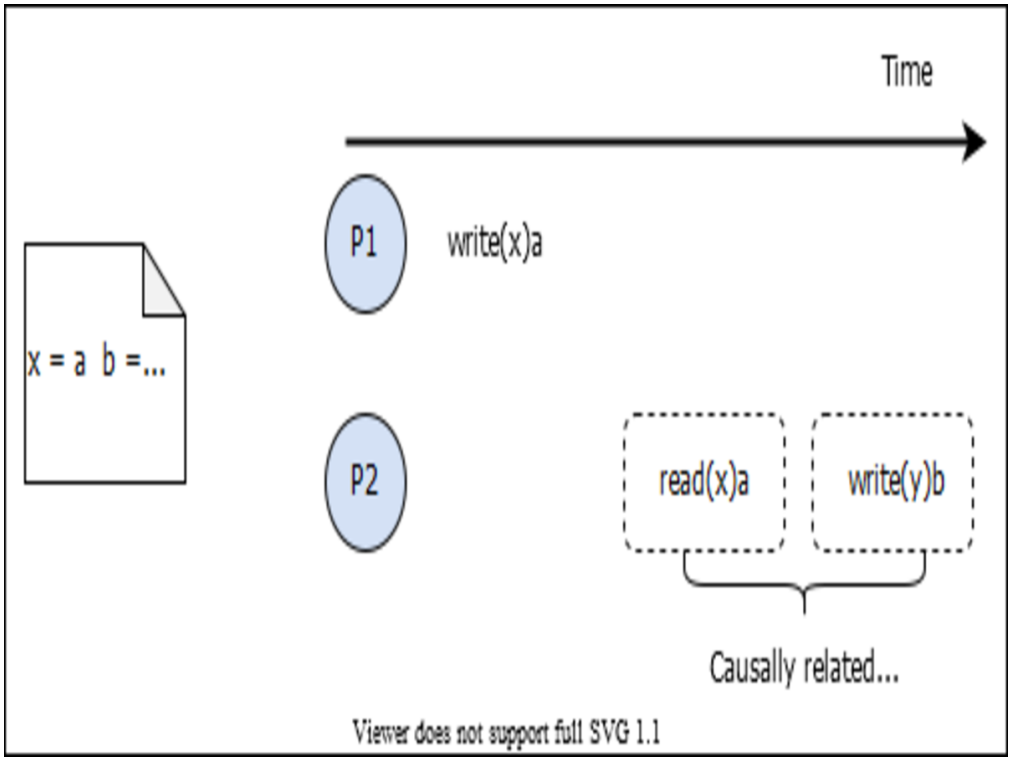


Figure 3.8: Causal consistency

This model doesn't ensure ordering for the operations that are not causally related. These operations can be seen in different possible orders.

Causal consistency is weaker overall, but stronger than the eventual consistency model. It's used to prevent non-intuitive behaviors.

Example

The causal consistency model is used in a commenting system. For example, for the replies to a comment on a Facebook post, we want to display comments after the comment it replies to. This is because there is a cause-and-effect relationship between a comment and its replies.

> **Note:** There are many consistency models other than the four discussed in this lesson, and there is still room for new consistency models. Researchers have developed new consistency models. For example, Wyatt Lloyd, et al.https://www.cs.prince-

ton.edu/~wlloyd/papers/cops-sosp11.pdf, proposed the **causal+consistency model** to speed up some specific types of transactions.

3.4 Sequential consistency

Sequential consistency is stronger than the causal consistency model. It preserves the ordering specified by each client's program. However, sequential consistency doesn't ensure that the writes are visible instantaneously or in the same order as they occurred according to some global clock.

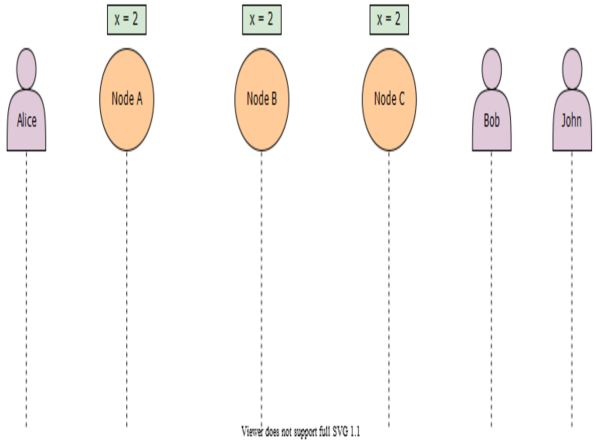
Example

In social networking applications, we usually don't care about the order in which some of our friends' posts appear. However, we still anticipate a single friend's posts to appear in the correct order in which they were created). Similarly , we expect our friends' comments in a post to display in the order that they were submitted. The sequential consistency model captures all of these qualities.

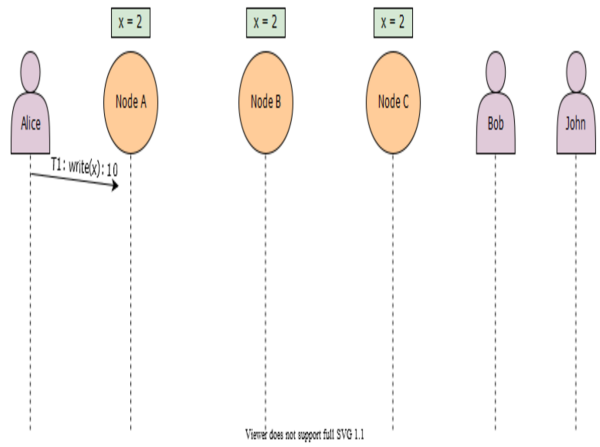
3.5 Strict consistency aka linearizability

A **strict consistency** or **linearizability** is the strongest consistency model. This model ensures that a read request from any replicas will get the latest write value. Once the client receives the acknowledgment that the write operation has been performed, other clients can read that value.

Linearizability is challenging to achieve in a distributed system. Some of the reasons for such challenges are variable network delays and failures. The following slides show depicts how variable network delays make it possible for different parties to see different values.

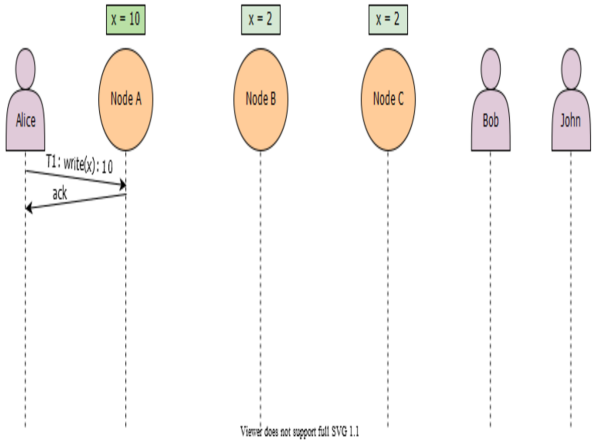


(a) There are three users of a system consisting of three nodes (replicas), and each node initially has a value of x that equals 2

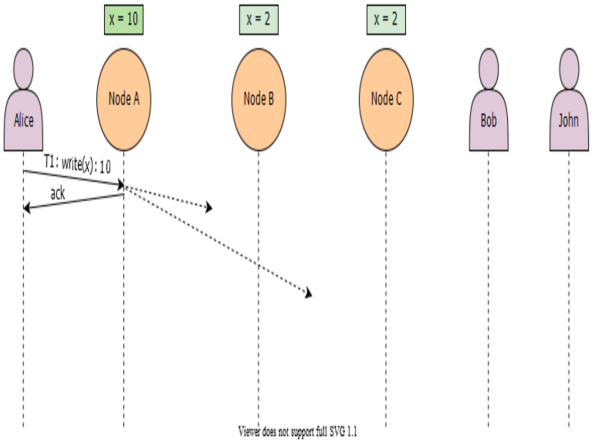


(b) Alice requests a write operation to change the value of x to 10

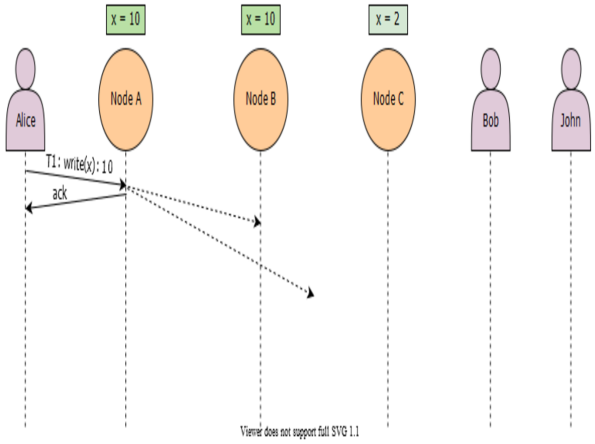
Usually, synchronous replication is one of the ingredients for achieving strong consistency, though it in itself is not sufficient. We might need consensus algorithms such as Paxos and Raft to achieve strong consistency.



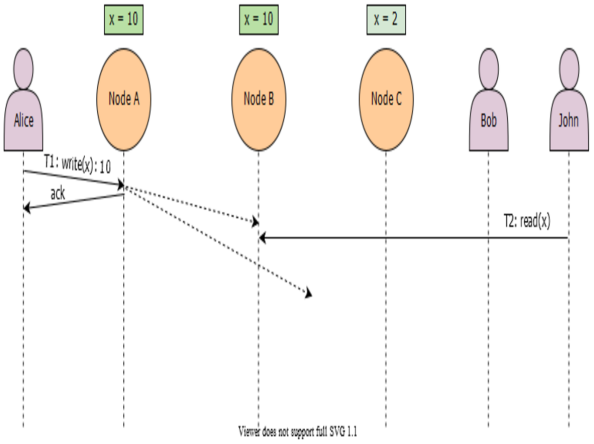
(a) Node A performs the write operation on x and returns an acknowledgment back to Alice



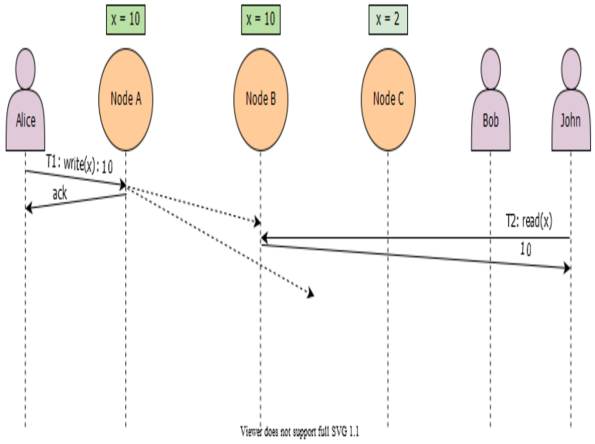
(b) Node A forwards the write operation to the other nodes in the system



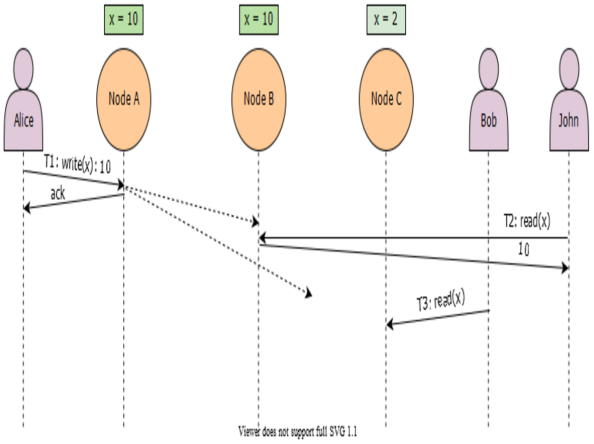
(a) Node B receives the write operation and changes the value of x to 10



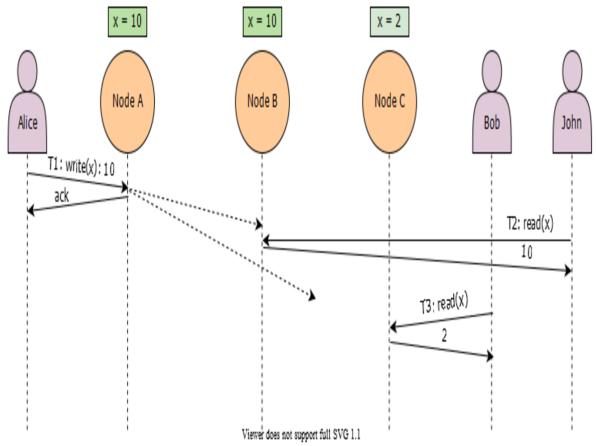
(b) John requests a read operation that is performed on Node B



(a) Node B has the latest value of x and returns it to John



(b) After John, Bob requests a read operation on x, which is performed on Node C



(a) Node C has not yet received the write operation, so it returns the old value of x = 2 to Bob

Linearizability affects the system's availability, which is why it's not always used. Applications with strong consistency requirements use techniques like **quorum-based replication** to increase the system's availability.

Example

Updating an account's password requires strict consistency. For example, if we suspect suspicious activity on our bank account, we immediately change our password so that no unauthorized users can access our account. If it were possible to access our account using an old password due to a lack of strict consistency, then changing passwords would be a useless security strategy.

> **Note:** Google's Spanner database claims to be linearizable for many of its operations.

3.6 Summary

Linearizable services appear to carry out transactions/operations in sequential, real-time order. They make it easier to create suitable applications on top of them by limiting the number of values that services can return to application processes.

Linearizable services have worse performance rates than services with weaker consistency in exchange for their strong assurances. Think about a read in a key-value store that returns the value written by a concurrent write. The read imposes no limits on future reads if the key-value store is weakly consistent.

Application programmers have to compromise performance and availability if they use services with strong consistency models. The models may break the invariants of applications built on top of them in exchange for increased performance.

4 The Spectrum of Failure Models

Failures are obvious in the world of distributed systems and can appear in various ways. They might come and go, or persist for a long period.

Failure models provide us a framework to reason about the impact of failures and possible ways to deal with them.

Here is an illustration that presents a spectrum of different failure models:

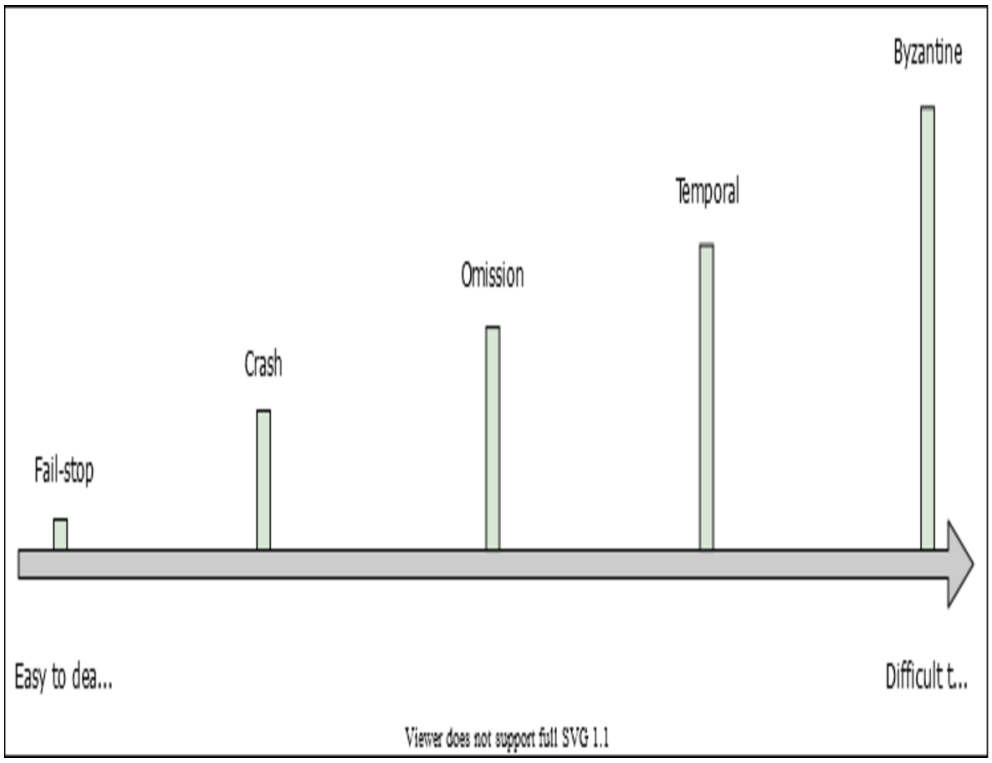


Figure 3.14: This is a spectrum of failure models. The difficulty level when dealing with a failure increases as we move to the right

4.1 Fail-stop

In this type of failure, a node in the distributed system halts permanently. However, the other nodes can still detect that node by communicating with it.

From the perspective of someone who builds distributed systems, fail-stop failures are the simplest and the most convenient.

4.2 Crash

In this type of failure, a node in the distributed system halts silently, and the other nodes can't detect that the node has stopped working.

4.3 Omission failures

In **omission failures**, the node fails to send or receive messages. There are two types of omission failures. If the node fails to respond to the incoming request, it's said to be a **send omission failure**. If the node fails to receive the request and thus can't acknowledge it, it's said to be a **receive omission failure**.

4.4 Temporal failures

In **temporal failures**, the node generates correct results, but is too late to be useful. This failure could be due to bad algorithms, a bad design strategy, or a loss of synchronization between the processor clocks.

4.5 Byzantine failures

In **Byzantine failures**, the node exhibits random behavior like transmitting arbitrary messages at arbitrary times, producing wrong results, or stopping midway. This mostly happens due to an attack by a malicious entity or a software bug. A byzantine failure is the most challenging type of failure to deal with.

CHAPTER 4

NON-FUNCTIONAL SYSTEM CHARACTERISTICS

1 Availability 28
2 Reliability 28
3 Scalability 28
4 Maintainability 29
5 Fault Tolerance 29
6 Let AI Evaluate Your Understanding of Non-Functional Requirements 29

Take a look at the key non-functional characteristics like availability, reliability, scalability, and fault tolerance in System Design.

1 Availability

Section content for "Availability" is being generated.

Section Details:

- Section ID: 5983424656965632
- Section Slug: availability
- Status: Content pending generation

Use the section content generation API to create the content for this section.

2 Reliability

Section content for "Reliability" is being generated.

Section Details:

- Section ID: 5173989319639040
- Section Slug: reliability
- Status: Content pending generation

Use the section content generation API to create the content for this section.

3 Scalability

Section content for "Scalability" is being generated.

Section Details:

- Section ID: 5991400583200768
- Section Slug: scalability
- Status: Content pending generation

Use the section content generation API to create the content for this section.

4 Maintainability

Section content for "Maintainability" is being generated.

Section Details:

- Section ID: 6540133490229248
- Section Slug: maintainability
- Status: Content pending generation

Use the section content generation API to create the content for this section.

5 Fault Tolerance

Section content for "Fault Tolerance" is being generated.

Section Details:

- Section ID: 6693927511392256
- Section Slug: fault-tolerance
- Status: Content pending generation

Use the section content generation API to create the content for this section.

6 Let AI Evaluate Your Understanding of Non-Functional Requirements

Section content for "Let AI Evaluate Your Understanding of Non-Functional Requirements" is being generated.

Section Details:

- Section ID: 6669534840487936
- Section Slug: let-ai-evaluate-your-understanding-of-non-functional-requirements
- Status: Content pending generation

Use the section content generation API to create the content for this section.

CHAPTER 5

BACK-OF-THE-ENVELOPE CALCULATIONS

1 Put Back-of-the-envelope Numbers in Perspective . . . 32
2 Examples of Resource Estimation 32

Learn how to estimate resource needs like servers, storage, and bandwidth using back-of-the-envelope calculations for System Design.

1 Put Back-of-the-envelope Numbers in Perspective

Section content for "Put Back-of-the-envelope Numbers in Perspective" is being generated.

Section Details:

- Section ID: 5711642666467328
- Section Slug: put-back-of-the-envelope-numbers-in-perspective
- Status: Content pending generation

Use the section content generation API to create the content for this section.

2 Examples of Resource Estimation

Section content for "Examples of Resource Estimation" is being generated.

Section Details:

- Section ID: 4766860129599488
- Section Slug: examples-of-resource-estimation
- Status: Content pending generation

Use the section content generation API to create the content for this section.

CHAPTER 6

BUILDING BLOCKS

1 Introduction to Building Blocks for Modern System Design 34

Explore the essential building blocks required for modern System Design, which will form the foundation for scalable systems later in the course.

1 Introduction to Building Blocks for Modern System Design

Section content for "Introduction to Building Blocks for Modern System Design" is being generated.

Section Details:

- Section ID: 6061247618875392
- Section Slug: introduction-to-building-blocks-for-modern-system-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

CHAPTER 7

DOMAIN NAME SYSTEM

1 Introduction to Domain Name System (DNS) 36
2 How the Domain Name System Works 36

Discover how DNS works and its role in System Design, including detailed insights into its design and functionality.

1 Introduction to Domain Name System (DNS)

Section content for "Introduction to Domain Name System (DNS)" is being generated.

Section Details:

- Section ID: 5728619204182016
- Section Slug: introduction-to-domain-name-system-dns
- Status: Content pending generation

Use the section content generation API to create the content for this section.

2 How the Domain Name System Works

Section content for "How the Domain Name System Works" is being generated.

Section Details:

- Section ID: 4570386150981632
- Section Slug: how-the-domain-name-system-works
- Status: Content pending generation

Use the section content generation API to create the content for this section.

CHAPTER 8

LOAD BALANCERS

1 Introduction to Load Balancers 38
2 Global and Local Load Balancing 38
3 Advanced Details of Load Balancers 38

Take a closer look at load balancers, their placement, and algorithms, including local and global load balancers and different tiers of load balancers.

1 Introduction to Load Balancers

Section content for "Introduction to Load Balancers" is being generated.

Section Details:

- Section ID: 4521972679049216
- Section Slug: introduction-to-load-balancers
- Status: Content pending generation

Use the section content generation API to create the content for this section.

2 Global and Local Load Balancing

Section content for "Global and Local Load Balancing" is being generated.

Section Details:

- Section ID: 5238392749293568
- Section Slug: global-and-local-load-balancing
- Status: Content pending generation

Use the section content generation API to create the content for this section.

3 Advanced Details of Load Balancers

Section content for "Advanced Details of Load Balancers" is being generated.

Section Details:

- Section ID: 6596709789990912
- Section Slug: advanced-details-of-load-balancers
- Status: Content pending generation

Use the section content generation API to create the content for this section.

CHAPTER 9

DATABASES

1 Introduction to Databases . 40
2 Types of Databases 40
3 Data Replication 40
4 Data Partitioning 41
5 Trade-offs in Databases . . . 41

Explore databases and their types, data replication, partitioning, and trade-offs essential for designing distributed systems.

1 Introduction to Databases

Section content for "Introduction to Databases" is being generated.

Section Details:

- Section ID: 4901035478351872
- Section Slug: introduction-to-databases
- Status: Content pending generation

Use the section content generation API to create the content for this section.

2 Types of Databases

Section content for "Types of Databases" is being generated.

Section Details:

- Section ID: 6521598450597888
- Section Slug: types-of-databases
- Status: Content pending generation

Use the section content generation API to create the content for this section.

3 Data Replication

Section content for "Data Replication" is being generated.

Section Details:

- Section ID: 5241733675220992
- Section Slug: data-replication
- Status: Content pending generation

Use the section content generation API to create the content for this section.

4 Data Partitioning

Section content for "Data Partitioning" is being generated.

Section Details:

- Section ID: 6254160546103296
- Section Slug: data-partitioning
- Status: Content pending generation

Use the section content generation API to create the content for this section.

5 Trade-offs in Databases

Section content for "Trade-offs in Databases" is being generated.

Section Details:

- Section ID: 5531655174881280
- Section Slug: trade-offs-in-databases
- Status: Content pending generation

Use the section content generation API to create the content for this section.

CHAPTER 10

KEY-VALUE STORE

1 System Design: The Key-value Store 44
2 Design of a Key-value Store 44
3 Ensure Scalability and Replication 44
4 Versioning Data and Achieving Configurability 45
5 Enable Fault Tolerance and Failure Detection 45

Learn how to design a scalable key-value store, focusing on replication, versioning, and fault tolerance.

1 System Design: The Key-value Store

Section content for "System Design: The Key-value Store" is being generated.

Section Details:

- Section ID: 4747701493432320
- Section Slug: system-design-the-key-value-store
- Status: Content pending generation

Use the section content generation API to create the content for this section.

2 Design of a Key-value Store

Section content for "Design of a Key-value Store" is being generated.

Section Details:

- Section ID: 5610927277211648
- Section Slug: design-of-a-key-value-store
- Status: Content pending generation

Use the section content generation API to create the content for this section.

3 Ensure Scalability and Replication

Section content for "Ensure Scalability and Replication" is being generated.

Section Details:

- Section ID: 5256346714243072
- Section Slug: ensure-scalability-and-replication
- Status: Content pending generation

Use the section content generation API to create the content for this section.

4 Versioning Data and Achieving Configurability

Section content for "Versioning Data and Achieving Configurability" is being generated.

Section Details:

- Section ID: 5979802812547072
- Section Slug: versioning-data-and-achieving-configurability
- Status: Content pending generation

Use the section content generation API to create the content for this section.

5 Enable Fault Tolerance and Failure Detection

Section content for "Enable Fault Tolerance and Failure Detection" is being generated.

Section Details:

- Section ID: 5842597234343936
- Section Slug: enable-fault-tolerance-and-failure-detection
- Status: Content pending generation

Use the section content generation API to create the content for this section.

CHAPTER 11

CONTENT DELIVERY NETWORK (CDN)

1 System Design: The Content Delivery Network (CDN) . . 48
2 Introduction to a CDN . . . 48
3 Design of a CDN 48
4 In-depth Investigation of CDN: Part 1 49
5 In-depth Investigation of CDN: Part 2 49
6 Evaluation of CDN's Design 49
7 Quiz on CDN's Design . . . 50

Discover CDN design, content caching strategies, consistency, and techniques for efficient content delivery.

1 System Design: The Content Delivery Network (CDN)

Section content for "System Design: The Content Delivery Network (CDN)" is being generated.

Section Details:

- Section ID: 6624266925899776
- Section Slug: system-design-the-content-delivery-network-cdn
- Status: Content pending generation

Use the section content generation API to create the content for this section.

2 Introduction to a CDN

Section content for "Introduction to a CDN" is being generated.

Section Details:

- Section ID: 5560114131501056
- Section Slug: introduction-to-a-cdn
- Status: Content pending generation

Use the section content generation API to create the content for this section.

3 Design of a CDN

Section content for "Design of a CDN" is being generated.

Section Details:

- Section ID: 5883973992972288
- Section Slug: design-of-a-cdn
- Status: Content pending generation

Use the section content generation API to create the content for this section.

4 In-depth Investigation of CDN: Part 1

Section content for "In-depth Investigation of CDN: Part 1" is being generated.

Section Details:

- Section ID: 5267936226312192
- Section Slug: in-depth-investigation-of-cdn-part-1
- Status: Content pending generation

Use the section content generation API to create the content for this section.

5 In-depth Investigation of CDN: Part 2

Section content for "In-depth Investigation of CDN: Part 2" is being generated.

Section Details:

- Section ID: 5382114350727168
- Section Slug: in-depth-investigation-of-cdn-part-2
- Status: Content pending generation

Use the section content generation API to create the content for this section.

6 Evaluation of CDN's Design

Section content for "Evaluation of CDN's Design" is being generated.

Section Details:

- Section ID: 5971383947624448
- Section Slug: evaluation-of-cdns-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

7 Quiz on CDN's Design

Section content for "Quiz on CDN's Design" is being generated.

Section Details:

- Section ID: 6624605451583488
- Section Slug: quiz-on-cdns-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

CHAPTER 12

SEQUENCER

1 System Design: Sequencer . 52
2 Design of a Unique ID Generator 52
3 Unique IDs with Causality . 52

Learn how to design a sequencer for generating unique IDs, focusing on causality and consistency in distributed systems.

1 System Design: Sequencer

Section content for "System Design: Sequencer" is being generated.

Section Details:

- Section ID: 6499939719053312
- Section Slug: system-design-sequencer
- Status: Content pending generation

Use the section content generation API to create the content for this section.

2 Design of a Unique ID Generator

Section content for "Design of a Unique ID Generator" is being generated.

Section Details:

- Section ID: 5216880444309504
- Section Slug: design-of-a-unique-id-generator
- Status: Content pending generation

Use the section content generation API to create the content for this section.

3 Unique IDs with Causality

Section content for "Unique IDs with Causality" is being generated.

Section Details:

- Section ID: 5836711686307840
- Section Slug: unique-ids-with-causality
- Status: Content pending generation

Use the section content generation API to create the content for this section.

CHAPTER 13

DISTRIBUTED MONITORING

1 System Design: Distributed Monitoring 54
2 Introduction to Distributed Monitoring 54
3 Prerequisites of a Monitoring System 54

Explore a distributed monitoring systems basics, types, and metrics for effective monitoring.

1 System Design: Distributed Monitoring

Section content for "System Design: Distributed Monitoring" is being generated.

Section Details:

- Section ID: 6310983387840512
- Section Slug: system-design-distributed-monitoring
- Status: Content pending generation

Use the section content generation API to create the content for this section.

2 Introduction to Distributed Monitoring

Section content for "Introduction to Distributed Monitoring" is being generated.

Section Details:

- Section ID: 4721220683038720
- Section Slug: introduction-to-distributed-monitoring
- Status: Content pending generation

Use the section content generation API to create the content for this section.

3 Prerequisites of a Monitoring System

Section content for "Prerequisites of a Monitoring System" is being generated.

Section Details:

- Section ID: 6071737539624960
- Section Slug: prerequisites-of-a-monitoring-system
- Status: Content pending generation

Use the section content generation API to create the content for this section.

CHAPTER 14

MONITOR SERVER-SIDE ERRORS

1 Design of a Monitoring System 56
2 Detailed Design of a Monitoring System 56
3 Visualize Data in a Monitoring System 56

Learn how to design a monitoring system and visualize it for tracking server-side errors in real-time.

1 Design of a Monitoring System

Section content for "Design of a Monitoring System" is being generated.

Section Details:

- Section ID: 5485445638520832
- Section Slug: design-of-a-monitoring-system
- Status: Content pending generation

Use the section content generation API to create the content for this section.

2 Detailed Design of a Monitoring System

Section content for "Detailed Design of a Monitoring System" is being generated.

Section Details:

- Section ID: 6718061809238016
- Section Slug: detailed-design-of-a-monitoring-system
- Status: Content pending generation

Use the section content generation API to create the content for this section.

3 Visualize Data in a Monitoring System

Section content for "Visualize Data in a Monitoring System" is being generated.

Section Details:

- Section ID: 5232629720285184
- Section Slug: visualize-data-in-a-monitoring-system
- Status: Content pending generation

Use the section content generation API to create the content for this section.

CHAPTER 15

MONITOR CLIENT-SIDE ERRORS

1 Focus on Client-side Errors in a Monitoring System 58
2 Design of a Client-side Monitoring System 58

Discover how to design a system to monitor client-side errors, ensuring robust and reliable applications.

1 Focus on Client-side Errors in a Monitoring System

Section content for "Focus on Client-side Errors in a Monitoring System" is being generated.

Section Details:

- Section ID: 4768611427680256
- Section Slug: focus-on-client-side-errors-in-a-monitoring-system
- Status: Content pending generation

Use the section content generation API to create the content for this section.

2 Design of a Client-side Monitoring System

Section content for "Design of a Client-side Monitoring System" is being generated.

Section Details:

- Section ID: 4769741293486080
- Section Slug: design-of-a-client-side-monitoring-system
- Status: Content pending generation

Use the section content generation API to create the content for this section.

CHAPTER 16

DISTRIBUTED CACHE

1 System Design: The Distributed Cache 60
2 Background of Distributed Cache 60
3 High-level Design of a Distributed Cache 60
4 Detailed Design of a Distributed Cache 61
5 Evaluation of a Distributed Cache's Design 61
6 Memcached versus Redis . . 61

Unpack the design of a distributed cache, focusing on high-level and detailed designs and evaluating its performance.

1 System Design: The Distributed Cache

Section content for "System Design: The Distributed Cache" is being generated.

Section Details:

- Section ID: 5053577315221504
- Section Slug: system-design-the-distributed-cache
- Status: Content pending generation

Use the section content generation API to create the content for this section.

2 Background of Distributed Cache

Section content for "Background of Distributed Cache" is being generated.

Section Details:

- Section ID: 5353674067148800
- Section Slug: background-of-distributed-cache
- Status: Content pending generation

Use the section content generation API to create the content for this section.

3 High-level Design of a Distributed Cache

Section content for "High-level Design of a Distributed Cache" is being generated.

Section Details:

- Section ID: 5103962818084864
- Section Slug: high-level-design-of-a-distributed-cache
- Status: Content pending generation

Use the section content generation API to create the content for this section.

4 Detailed Design of a Distributed Cache

Section content for "Detailed Design of a Distributed Cache" is being generated.

Section Details:

- Section ID: 4731033765543936
- Section Slug: detailed-design-of-a-distributed-cache
- Status: Content pending generation

Use the section content generation API to create the content for this section.

5 Evaluation of a Distributed Cache's Design

Section content for "Evaluation of a Distributed Cache's Design" is being generated.

Section Details:

- Section ID: 6094096892690432
- Section Slug: evaluation-of-a-distributed-caches-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

6 Memcached versus Redis

Section content for "Memcached versus Redis" is being generated.

Section Details:

- Section ID: 5441771256938496
- Section Slug: memcached-versus-redis
- Status: Content pending generation

Use the section content generation API to create the content for this section.

CHAPTER 17

DISTRIBUTED MESSAGING QUEUE

1 System Design: The Distributed Messaging Queue . . 64
2 Requirements of a Distributed Messaging Queues Design 64
3 Considerations of a Distributed Messaging Queues Design 64
4 Design of a Distributed Messaging Queue: Part 1 65
5 Design of a Distributed Messaging Queue: Part 2 65
6 Evaluation of a Distributed Messaging Queues Design . . 65
7 Quiz on the Distributed Messaging Queues Design 66

Examine the design of a distributed messaging queue, addressing requirements, design considerations, and performance evaluations.

1 System Design: The Distributed Messaging Queue

Section content for "System Design: The Distributed Messaging Queue" is being generated.

Section Details:

- Section ID: 5148400467312640
- Section Slug: system-design-the-distributed-messaging-queue
- Status: Content pending generation

Use the section content generation API to create the content for this section.

2 Requirements of a Distributed Messaging Queues Design

Section content for "Requirements of a Distributed Messaging Queues Design" is being generated.

Section Details:

- Section ID: 5806944861814784
- Section Slug: requirements-of-a-distributed-messaging-queues-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

3 Considerations of a Distributed Messaging Queues Design

Section content for "Considerations of a Distributed Messaging Queues Design" is being generated.

Section Details:

- Section ID: 6191662786674688
- Section Slug: considerations-of-a-distributed-messaging-queues-design

- Status: Content pending generation

Use the section content generation API to create the content for this section.

4 Design of a Distributed Messaging Queue: Part 1

Section content for "Design of a Distributed Messaging Queue: Part 1" is being generated.

Section Details:

- Section ID: 6330012742975488
- Section Slug: design-of-a-distributed-messaging-queue-part-1
- Status: Content pending generation

Use the section content generation API to create the content for this section.

5 Design of a Distributed Messaging Queue: Part 2

Section content for "Design of a Distributed Messaging Queue: Part 2" is being generated.

Section Details:

- Section ID: 4953511267139584
- Section Slug: design-of-a-distributed-messaging-queue-part-2
- Status: Content pending generation

Use the section content generation API to create the content for this section.

6 Evaluation of a Distributed Messaging Queues Design

Section content for "Evaluation of a Distributed Messaging Queues Design" is being generated.

Section Details:

- Section ID: 4654974423793664

- Section Slug: evaluation-of-a-distributed-messaging-queues-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

7 Quiz on the Distributed Messaging Queues Design

Section content for "Quiz on the Distributed Messaging Queues Design" is being generated.

Section Details:

- Section ID: 5261732032806912
- Section Slug: quiz-on-the-distributed-messaging-queues-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

CHAPTER 18

PUB-SUB

1 System Design: The Pub-sub Abstraction 68
2 Introduction to Pub-sub . . 68
3 Design of a Pub-sub System 68

Learn the design of a pub-sub system, focusing on how to implement and optimize this communication pattern in distributed systems.

1 System Design: The Pub-sub Abstraction

Section content for "System Design: The Pub-sub Abstraction" is being generated.

Section Details:

- Section ID: 4996814243889152
- Section Slug: system-design-the-pub-sub-abstraction
- Status: Content pending generation

Use the section content generation API to create the content for this section.

2 Introduction to Pub-sub

Section content for "Introduction to Pub-sub" is being generated.

Section Details:

- Section ID: 5155663642099712
- Section Slug: introduction-to-pub-sub
- Status: Content pending generation

Use the section content generation API to create the content for this section.

3 Design of a Pub-sub System

Section content for "Design of a Pub-sub System" is being generated.

Section Details:

- Section ID: 5367894619193344
- Section Slug: design-of-a-pub-sub-system
- Status: Content pending generation

Use the section content generation API to create the content for this section.

CHAPTER 19

RATE LIMITER

1 System Design: The Rate Limiter 70
2 Requirements of a Rate Limiters Design 70
3 Design of a Rate Limiter . . 70
4 Rate Limiter Algorithms . . 71
5 Quiz on the Rate Limiters Design 71

Explore the design of a rate limiter, covering essential algorithms and considerations for managing traffic and ensuring system stability.

1 System Design: The Rate Limiter

Section content for "System Design: The Rate Limiter" is being generated.

Section Details:

- Section ID: 4770834422169600
- Section Slug: system-design-the-rate-limiter
- Status: Content pending generation

Use the section content generation API to create the content for this section.

2 Requirements of a Rate Limiters Design

Section content for "Requirements of a Rate Limiters Design" is being generated.

Section Details:

- Section ID: 5887506125422592
- Section Slug: requirements-of-a-rate-limiters-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

3 Design of a Rate Limiter

Section content for "Design of a Rate Limiter" is being generated.

Section Details:

- Section ID: 5287353329647616
- Section Slug: design-of-a-rate-limiter
- Status: Content pending generation

Use the section content generation API to create the content for this section.

4 Rate Limiter Algorithms

Section content for "Rate Limiter Algorithms" is being generated.

Section Details:

- Section ID: 5447913559293952
- Section Slug: rate-limiter-algorithms
- Status: Content pending generation

Use the section content generation API to create the content for this section.

5 Quiz on the Rate Limiters Design

Section content for "Quiz on the Rate Limiters Design" is being generated.

Section Details:

- Section ID: 6564352194248704
- Section Slug: quiz-on-the-rate-limiters-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

CHAPTER 20

BLOB STORE

1 System Design: A Blob Store 74
2 Requirements of a Blob Store's Design 74
3 Design of a Blob Store . . . 74
4 Design Considerations of a Blob Store 75
5 Evaluation of a Blob Store's Design 75
6 Quiz on the Blob Store's Design 75

Learn how to design a blob store, focusing on scalability, storage, and performance evaluation.

1 System Design: A Blob Store

Section content for "System Design: A Blob Store" is being generated.

Section Details:

- Section ID: 4862646238576640
- Section Slug: system-design-a-blob-store
- Status: Content pending generation

Use the section content generation API to create the content for this section.

2 Requirements of a Blob Store's Design

Section content for "Requirements of a Blob Store's Design" is being generated.

Section Details:

- Section ID: 5129194917068800
- Section Slug: requirements-of-a-blob-stores-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

3 Design of a Blob Store

Section content for "Design of a Blob Store" is being generated.

Section Details:

- Section ID: 4910790825476096
- Section Slug: design-of-a-blob-store
- Status: Content pending generation

Use the section content generation API to create the content for this section.

4 Design Considerations of a Blob Store

Section content for "Design Considerations of a Blob Store" is being generated.

Section Details:

- Section ID: 4593802865541120
- Section Slug: design-considerations-of-a-blob-store
- Status: Content pending generation

Use the section content generation API to create the content for this section.

5 Evaluation of a Blob Store's Design

Section content for "Evaluation of a Blob Store's Design" is being generated.

Section Details:

- Section ID: 5348783000387584
- Section Slug: evaluation-of-a-blob-stores-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

6 Quiz on the Blob Store's Design

Section content for "Quiz on the Blob Store's Design" is being generated.

Section Details:

- Section ID: 6060039001604096
- Section Slug: quiz-on-the-blob-stores-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

CHAPTER 21

DISTRIBUTED SEARCH

1 System Design: The Distributed Search 78
2 Requirements of a Distributed Search System's Design 78
3 Indexing in a Distributed Search 78
4 Design of a Distributed Search 79
5 Scaling Search and Indexing 79
6 Evaluation of a Distributed Search's Design 79

Step through distributed search design in stages, with a special focus on indexing, scaling, and replication.

1 System Design: The Distributed Search

Section content for "System Design: The Distributed Search" is being generated.

Section Details:

- Section ID: 5400897294696448
- Section Slug: system-design-the-distributed-search
- Status: Content pending generation

Use the section content generation API to create the content for this section.

2 Requirements of a Distributed Search System's Design

Section content for "Requirements of a Distributed Search System's Design" is being generated.

Section Details:

- Section ID: 5706146547761152
- Section Slug: requirements-of-a-distributed-search-systems-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

3 Indexing in a Distributed Search

Section content for "Indexing in a Distributed Search" is being generated.

Section Details:

- Section ID: 6595902341120000
- Section Slug: indexing-in-a-distributed-search
- Status: Content pending generation

Use the section content generation API to create the content for this section.

4 Design of a Distributed Search

Section content for "Design of a Distributed Search" is being generated.

Section Details:

- Section ID: 6573171574046720
- Section Slug: design-of-a-distributed-search
- Status: Content pending generation

Use the section content generation API to create the content for this section.

5 Scaling Search and Indexing

Section content for "Scaling Search and Indexing" is being generated.

Section Details:

- Section ID: 5899429382455296
- Section Slug: scaling-search-and-indexing
- Status: Content pending generation

Use the section content generation API to create the content for this section.

6 Evaluation of a Distributed Search's Design

Section content for "Evaluation of a Distributed Search's Design" is being generated.

Section Details:

- Section ID: 4962799423324160
- Section Slug: evaluation-of-a-distributed-searchs-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

CHAPTER 22

DISTRIBUTED LOGGING

1 System Design: Distributed Logging 82
2 Introduction to Distributed Logging 82
3 Design of a Distributed Logging Service 82

Understand the importance and design of a distributed logging service, emphasizing capturing and analyzing logs across distributed systems.

1 System Design: Distributed Logging

Section content for "System Design: Distributed Logging" is being generated.

Section Details:

- Section ID: 6747547922333696
- Section Slug: system-design-distributed-logging
- Status: Content pending generation

Use the section content generation API to create the content for this section.

2 Introduction to Distributed Logging

Section content for "Introduction to Distributed Logging" is being generated.

Section Details:

- Section ID: 5621964999229440
- Section Slug: introduction-to-distributed-logging
- Status: Content pending generation

Use the section content generation API to create the content for this section.

3 Design of a Distributed Logging Service

Section content for "Design of a Distributed Logging Service" is being generated.

Section Details:

- Section ID: 4835612456124416
- Section Slug: design-of-a-distributed-logging-service
- Status: Content pending generation

Use the section content generation API to create the content for this section.

CHAPTER 23

DISTRIBUTED TASK SCHEDULER

1 System Design: The Distributed Task Scheduler . . . 84
2 Requirements of a Distributed Task Scheduler's Design 84
3 Design of a Distributed Task Scheduler 84
4 Design Considerations of a Distributed Task Scheduler . 85
5 Evaluation of a Distributed Task Scheduler's Design . . . 85

Examine the design of a task scheduler addressing issues like prioritizing, task idempotency, queuing, and resource capacity optimization.

1 System Design: The Distributed Task Scheduler

Section content for "System Design: The Distributed Task Scheduler" is being generated.

Section Details:

- Section ID: 6152021643624448
- Section Slug: system-design-the-distributed-task-scheduler
- Status: Content pending generation

Use the section content generation API to create the content for this section.

2 Requirements of a Distributed Task Scheduler's Design

Section content for "Requirements of a Distributed Task Scheduler's Design" is being generated.

Section Details:

- Section ID: 6681764793942016
- Section Slug: requirements-of-a-distributed-task-schedulers-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

3 Design of a Distributed Task Scheduler

Section content for "Design of a Distributed Task Scheduler" is being generated.

Section Details:

- Section ID: 4969977353863168
- Section Slug: design-of-a-distributed-task-scheduler
- Status: Content pending generation

Use the section content generation API to create the content for this section.

4 Design Considerations of a Distributed Task Scheduler

Section content for "Design Considerations of a Distributed Task Scheduler" is being generated.

Section Details:

- Section ID: 4856850956156928
- Section Slug: design-considerations-of-a-distributed-task-scheduler
- Status: Content pending generation

Use the section content generation API to create the content for this section.

5 Evaluation of a Distributed Task Scheduler's Design

Section content for "Evaluation of a Distributed Task Scheduler's Design" is being generated.

Section Details:

- Section ID: 5231625754902528
- Section Slug: evaluation-of-a-distributed-task-schedulers-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

CHAPTER 24

SHARDED COUNTERS

1 System Design: The Sharded Counters 88
2 High-level Design of Sharded Counters 88
3 Detailed Design of Sharded Counters 88
4 Quiz on the Sharded Counters' Design 89

Get familiar with sharded counters and their importance, starting from high-level design and then pivoting to detailed System Design.

1 System Design: The Sharded Counters

Section content for "System Design: The Sharded Counters" is being generated.

Section Details:

- Section ID: 6071347163955200
- Section Slug: system-design-the-sharded-counters
- Status: Content pending generation

Use the section content generation API to create the content for this section.

2 High-level Design of Sharded Counters

Section content for "High-level Design of Sharded Counters" is being generated.

Section Details:

- Section ID: 6126387596886016
- Section Slug: high-level-design-of-sharded-counters
- Status: Content pending generation

Use the section content generation API to create the content for this section.

3 Detailed Design of Sharded Counters

Section content for "Detailed Design of Sharded Counters" is being generated.

Section Details:

- Section ID: 5076411916484608
- Section Slug: detailed-design-of-sharded-counters
- Status: Content pending generation

Use the section content generation API to create the content for this section.

4 Quiz on the Sharded Counters' Design

Section content for "Quiz on the Sharded Counters' Design" is being generated.

Section Details:

- Section ID: 5659634509021184
- Section Slug: quiz-on-the-sharded-counters-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

CHAPTER 25

CONCLUDING THE BUILDING BLOCKS DISCUSSION

1 AI Evaluation of Building Blocks in E-Commerce Platform 92
2 AI Evaluation of Building Blocks in Online Education System 92
3 Wrapping Up the Building Blocks Discussion 92
4 The RESHADED Approach for System Design 93

Wrap up the discussion on building blocks, evaluate your understanding using AI, and learn the RESHADED approach to tackle unseen System Design problems.

1 AI Evaluation of Building Blocks in E-Commerce Platform

Section content for "AI Evaluation of Building Blocks in E-Commerce Platform" is being generated.

Section Details:

- Section ID: 5469233437868032
- Section Slug: ai-evaluation-of-building-blocks-in-e-commerce-platform
- Status: Content pending generation

Use the section content generation API to create the content for this section.

2 AI Evaluation of Building Blocks in Online Education System

Section content for "AI Evaluation of Building Blocks in Online Education System" is being generated.

Section Details:

- Section ID: 6068850835718144
- Section Slug: ai-evaluation-of-building-blocks-in-online-education-system
- Status: Content pending generation

Use the section content generation API to create the content for this section.

3 Wrapping Up the Building Blocks Discussion

Section content for "Wrapping Up the Building Blocks Discussion" is being generated.

Section Details:

- Section ID: 4682017479852032
- Section Slug: wrapping-up-the-building-blocks-discussion

- Status: Content pending generation

Use the section content generation API to create the content for this section.

4 The RESHADED Approach for System Design

Section content for "The RESHADED Approach for System Design" is being generated.

Section Details:

- Section ID: 4564800009142272
- Section Slug: the-reshaded-approach-for-system-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

CHAPTER 26

DESIGN YOUTUBE

1 System Design: YouTube . . 96
2 Requirements of YouTube's Design 99
3 Design of YouTube 107
4 Evaluation of YouTube's Design 115
5 The Reality Is More Complicated 120
6 Quiz on YouTube's Design . 129

Learn YouTube System Design, starting with requirements, high-level and detailed design, evaluation of the design, and handling real-world complexities.

1 System Design: YouTube

Section content for "System Design: YouTube" is being generated.

Section Details:

- Section ID: 5360542734090240
- Section Slug: system-design-youtube
- Status: Content pending generation

Use the section content generation API to create the content for this section.

2 Requirements of YouTube's Design

Section content for "Requirements of YouTube's Design" is being generated.

Section Details:

- Section ID: 5574324458618880
- Section Slug: requirements-of-youtubes-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

3 Design of YouTube

Section content for "Design of YouTube" is being generated.

Section Details:

- Section ID: 5334361548783616
- Section Slug: design-of-youtube
- Status: Content pending generation

Use the section content generation API to create the content for this section.

4 Evaluation of YouTube's Design

Section content for "Evaluation of YouTube's Design" is being generated.

Section Details:

- Section ID: 4932171382390784
- Section Slug: evaluation-of-youtubes-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

5 The Reality Is More Complicated

Section content for "The Reality Is More Complicated" is being generated.

Section Details:

- Section ID: 5471711817891840
- Section Slug: the-reality-is-more-complicated
- Status: Content pending generation

Use the section content generation API to create the content for this section.

6 Quiz on YouTube's Design

Section content for "Quiz on YouTube's Design" is being generated.

Section Details:

- Section ID: 6669179066777600
- Section Slug: quiz-on-youtubes-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

CHAPTER 27

DESIGN QUORA

1 System Design: Quora . . . 134
2 Requirements of Quora's Design 134
3 Initial Design of Quora . . . 134
4 Final Design of Quora . . . 135
5 Evaluation of Quoras Design 135

Explore the System Design of Quora incrementally by starting with key requirements and challenges in building a scalable Q\A platform.

1 System Design: Quora

Section content for "System Design: Quora" is being generated.

Section Details:

- Section ID: 5132399700869120
- Section Slug: system-design-quora
- Status: Content pending generation

Use the section content generation API to create the content for this section.

2 Requirements of Quora's Design

Section content for "Requirements of Quora's Design" is being generated.

Section Details:

- Section ID: 6254998790340608
- Section Slug: requirements-of-quoras-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

3 Initial Design of Quora

Section content for "Initial Design of Quora" is being generated.

Section Details:

- Section ID: 6337842040537088
- Section Slug: initial-design-of-quora
- Status: Content pending generation

Use the section content generation API to create the content for this section.

4 Final Design of Quora

Section content for "Final Design of Quora" is being generated.

Section Details:

- Section ID: 6323245837451264
- Section Slug: final-design-of-quora
- Status: Content pending generation

Use the section content generation API to create the content for this section.

5 Evaluation of Quoras Design

Section content for "Evaluation of Quoras Design" is being generated.

Section Details:

- Section ID: 5580185650331648
- Section Slug: evaluation-of-quoras-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

CHAPTER 28

DESIGN GOOGLE MAPS

1 System Design: Google Maps 138
2 Requirements of Google Maps' Design 138
3 Design of Google Maps . . . 138
4 Challenges of Google Maps' Design 139
5 Detailed Design of Google Maps 139
6 Evaluation of Google Maps' Design 139

Walk through the System Design of Google Maps, focusing on API design, scalability, finding optimal routes, and ETA computation.

1 System Design: Google Maps

Section content for "System Design: Google Maps" is being generated.

Section Details:

- Section ID: 5138720050642944
- Section Slug: system-design-google-maps
- Status: Content pending generation

Use the section content generation API to create the content for this section.

2 Requirements of Google Maps' Design

Section content for "Requirements of Google Maps' Design" is being generated.

Section Details:

- Section ID: 5704949308850176
- Section Slug: requirements-of-google-maps-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

3 Design of Google Maps

Section content for "Design of Google Maps" is being generated.

Section Details:

- Section ID: 6659265711833088
- Section Slug: design-of-google-maps
- Status: Content pending generation

Use the section content generation API to create the content for this section.

4 Challenges of Google Maps' Design

Section content for "Challenges of Google Maps' Design" is being generated.

Section Details:

- Section ID: 4659727862071296
- Section Slug: challenges-of-google-maps-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

5 Detailed Design of Google Maps

Section content for "Detailed Design of Google Maps" is being generated.

Section Details:

- Section ID: 6026783136940032
- Section Slug: detailed-design-of-google-maps
- Status: Content pending generation

Use the section content generation API to create the content for this section.

6 Evaluation of Google Maps' Design

Section content for "Evaluation of Google Maps' Design" is being generated.

Section Details:

- Section ID: 6335210467098624
- Section Slug: evaluation-of-google-maps-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

CHAPTER 29

DESIGN A PROXIMITY SERVICE / YELP

1 System Design: Yelp 142
2 Requirements of Yelps Design 142
3 Design of Yelp 142
4 Design Considerations of Yelp 143
5 Quiz on Yelp's Design 143

Take a closer look at the System Design of a proximity service like Yelp, addressing requirements like searching, scaling, and dynamic segments.

1 System Design: Yelp

Section content for "System Design: Yelp" is being generated.

Section Details:

- Section ID: 5264443638546432
- Section Slug: system-design-yelp
- Status: Content pending generation

Use the section content generation API to create the content for this section.

2 Requirements of Yelps Design

Section content for "Requirements of Yelps Design" is being generated.

Section Details:

- Section ID: 4552284171665408
- Section Slug: requirements-of-yelps-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

3 Design of Yelp

Section content for "Design of Yelp" is being generated.

Section Details:

- Section ID: 5027439944400896
- Section Slug: design-of-yelp
- Status: Content pending generation

Use the section content generation API to create the content for this section.

4 Design Considerations of Yelp

Section content for "Design Considerations of Yelp" is being generated.

Section Details:

- Section ID: 5518783229198336
- Section Slug: design-considerations-of-yelp
- Status: Content pending generation

Use the section content generation API to create the content for this section.

5 Quiz on Yelp's Design

Section content for "Quiz on Yelp's Design" is being generated.

Section Details:

- Section ID: 6678825089368064
- Section Slug: quiz-on-yelps-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

CHAPTER 30

DESIGN UBER

1 System Design: Uber 146
2 Requirements of Ubers Design 146
3 High-level Design of Uber . . 146
4 Detailed Design of Uber . . 147
5 Payment Service and Fraud Detection in Uber Design . . 147
6 Evaluation of Ubers Design . 147
7 Quiz on Uber's Design . . . 148

Understand how to design Uber, address requirements for ride-sharing platforms, detailed design, and fraud detection.

1 System Design: Uber

Section content for "System Design: Uber" is being generated.

Section Details:

- Section ID: 6344806557286400
- Section Slug: system-design-uber
- Status: Content pending generation

Use the section content generation API to create the content for this section.

2 Requirements of Ubers Design

Section content for "Requirements of Ubers Design" is being generated.

Section Details:

- Section ID: 6376322179268608
- Section Slug: requirements-of-ubers-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

3 High-level Design of Uber

Section content for "High-level Design of Uber" is being generated.

Section Details:

- Section ID: 5429357153091584
- Section Slug: high-level-design-of-uber
- Status: Content pending generation

Use the section content generation API to create the content for this section.

4 Detailed Design of Uber

Section content for "Detailed Design of Uber" is being generated.

Section Details:

- Section ID: 6683135341494272
- Section Slug: detailed-design-of-uber
- Status: Content pending generation

Use the section content generation API to create the content for this section.

5 Payment Service and Fraud Detection in Uber Design

Section content for "Payment Service and Fraud Detection in Uber Design" is being generated.

Section Details:

- Section ID: 6353616188735488
- Section Slug: payment-service-and-fraud-detection-in-uber-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

6 Evaluation of Ubers Design

Section content for "Evaluation of Ubers Design" is being generated.

Section Details:

- Section ID: 5469445371199488
- Section Slug: evaluation-of-ubers-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

7 Quiz on Uber's Design

Section content for "Quiz on Uber's Design" is being generated.

Section Details:

- Section ID: 6590232799739904
- Section Slug: quiz-on-ubers-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

CHAPTER 31

DESIGN TWITTER

1 System Design: Twitter . . . 150
2 Requirements of Twitters Design 150
3 High-level Design of Twitter 150
4 Detailed Design of Twitter . 151
5 Client-side Load Balancer for Twitter 151
6 Quiz on Twitter's Design . . 151

Learn Twitter System Design, covering aspects like user interaction, API design, caching, storage, and client-side load balancing.

1 System Design: Twitter

Section content for "System Design: Twitter" is being generated.

Section Details:

- Section ID: 4687773419700224
- Section Slug: system-design-twitter
- Status: Content pending generation

Use the section content generation API to create the content for this section.

2 Requirements of Twitters Design

Section content for "Requirements of Twitters Design" is being generated.

Section Details:

- Section ID: 5920537112477696
- Section Slug: requirements-of-twitters-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

3 High-level Design of Twitter

Section content for "High-level Design of Twitter" is being generated.

Section Details:

- Section ID: 6194554750894080
- Section Slug: high-level-design-of-twitter
- Status: Content pending generation

Use the section content generation API to create the content for this section.

4 Detailed Design of Twitter

Section content for "Detailed Design of Twitter" is being generated.

Section Details:

- Section ID: 6156135136755712
- Section Slug: detailed-design-of-twitter
- Status: Content pending generation

Use the section content generation API to create the content for this section.

5 Client-side Load Balancer for Twitter

Section content for "Client-side Load Balancer for Twitter" is being generated.

Section Details:

- Section ID: 5379128533975040
- Section Slug: client-side-load-balancer-for-twitter
- Status: Content pending generation

Use the section content generation API to create the content for this section.

6 Quiz on Twitter's Design

Section content for "Quiz on Twitter's Design" is being generated.

Section Details:

- Section ID: 6740284133605376
- Section Slug: quiz-on-twitters-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

CHAPTER 32

DESIGN NEWSFEED SYSTEM

1 System Design: Newsfeed System 154
2 Requirements of a Newsfeed Systems Design 154
3 Design of a Newsfeed System 154
4 Evaluation of a Newsfeed Systems Design 155

Master newsfeed System Design, covering aspects like functional and non-functional requirements, storage schemas, newsfeed generation, and publishing.

1 System Design: Newsfeed System

Section content for "System Design: Newsfeed System" is being generated.

Section Details:

- Section ID: 4943636425080832
- Section Slug: system-design-newsfeed-system
- Status: Content pending generation

Use the section content generation API to create the content for this section.

2 Requirements of a Newsfeed Systems Design

Section content for "Requirements of a Newsfeed Systems Design" is being generated.

Section Details:

- Section ID: 5590893725220864
- Section Slug: requirements-of-a-newsfeed-systems-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

3 Design of a Newsfeed System

Section content for "Design of a Newsfeed System" is being generated.

Section Details:

- Section ID: 5697004751028224
- Section Slug: design-of-a-newsfeed-system
- Status: Content pending generation

Use the section content generation API to create the content for this section.

4 Evaluation of a Newsfeed Systems Design

Section content for "Evaluation of a Newsfeed Systems Design" is being generated.

Section Details:

- Section ID: 5660564772093952
- Section Slug: evaluation-of-a-newsfeed-systems-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

CHAPTER 33

DESIGN INSTAGRAM

1 System Design: Instagram . 158
2 Requirements of Instagrams Design 158
3 Design of Instagram 158
4 Detailed Design of Instagram 159
5 Quiz on Instagrams Design . 159

Explore Instagrams System Design, covering API design, storage schema, and timeline generation using pull, push, and hybrid approaches.

1 System Design: Instagram

Section content for "System Design: Instagram" is being generated.

Section Details:

- Section ID: 5206574926921728
- Section Slug: system-design-instagram
- Status: Content pending generation

Use the section content generation API to create the content for this section.

2 Requirements of Instagrams Design

Section content for "Requirements of Instagrams Design" is being generated.

Section Details:

- Section ID: 5571545748996096
- Section Slug: requirements-of-instagrams-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

3 Design of Instagram

Section content for "Design of Instagram" is being generated.

Section Details:

- Section ID: 5160831733465088
- Section Slug: design-of-instagram
- Status: Content pending generation

Use the section content generation API to create the content for this section.

4 Detailed Design of Instagram

Section content for "Detailed Design of Instagram" is being generated.

Section Details:

- Section ID: 6638179708305408
- Section Slug: detailed-design-of-instagram
- Status: Content pending generation

Use the section content generation API to create the content for this section.

5 Quiz on Instagrams Design

Section content for "Quiz on Instagrams Design" is being generated.

Section Details:

- Section ID: 5987143142342656
- Section Slug: quiz-on-instagrams-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

CHAPTER 34

DESIGN A URL SHORTENING SERVICE / TINYURL

1 System Design: TinyURL . 162
2 Requirements of TinyURL's Design 162
3 Design and Deployment of TinyURL 162
4 Encoder for TinyURL 163
5 Evaluation of TinyURL's Design 163
6 Quiz on TinyURL's Design . 163

Decode the System Design of a URL shortening service like TinyURL, emphasizing requirements like encoding, scalability, and high readability.

1 System Design: TinyURL

Section content for "System Design: TinyURL" is being generated.

Section Details:

- Section ID: 6627189651144704
- Section Slug: system-design-tinyurl
- Status: Content pending generation

Use the section content generation API to create the content for this section.

2 Requirements of TinyURL's Design

Section content for "Requirements of TinyURL's Design" is being generated.

Section Details:

- Section ID: 5146211836755968
- Section Slug: requirements-of-tinyurls-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

3 Design and Deployment of TinyURL

Section content for "Design and Deployment of TinyURL" is being generated.

Section Details:

- Section ID: 5753981720068096
- Section Slug: design-and-deployment-of-tinyurl
- Status: Content pending generation

Use the section content generation API to create the content for this section.

4 Encoder for TinyURL

Section content for "Encoder for TinyURL" is being generated.

Section Details:

- Section ID: 5084162431647744
- Section Slug: encoder-for-tinyurl
- Status: Content pending generation

Use the section content generation API to create the content for this section.

5 Evaluation of TinyURL's Design

Section content for "Evaluation of TinyURL's Design" is being generated.

Section Details:

- Section ID: 6451282881019904
- Section Slug: evaluation-of-tinyurls-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

6 Quiz on TinyURL's Design

Section content for "Quiz on TinyURL's Design" is being generated.

Section Details:

- Section ID: 6050864745152512
- Section Slug: quiz-on-tinyurls-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

CHAPTER 35

DESIGN A WEB CRAWLER

1 System Design: Web Crawler 166
2 Requirements of a Web Crawler's Design 166
3 Design of a Web Crawler . . 166
4 Design Improvements of a Web Crawler 167
5 Evaluation of Web Crawler's Design 167

Explore the System Design of a web crawler, including its key components, such as a crawler, scheduler, HTML fetcher, storage, and crawling traps handler.

1 System Design: Web Crawler

Section content for "System Design: Web Crawler" is being generated.

Section Details:

- Section ID: 4695113376989184
- Section Slug: system-design-web-crawler
- Status: Content pending generation

Use the section content generation API to create the content for this section.

2 Requirements of a Web Crawler's Design

Section content for "Requirements of a Web Crawler's Design" is being generated.

Section Details:

- Section ID: 6454882160082944
- Section Slug: requirements-of-a-web-crawlers-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

3 Design of a Web Crawler

Section content for "Design of a Web Crawler" is being generated.

Section Details:

- Section ID: 6632099624255488
- Section Slug: design-of-a-web-crawler
- Status: Content pending generation

Use the section content generation API to create the content for this section.

4 Design Improvements of a Web Crawler

Section content for "Design Improvements of a Web Crawler" is being generated.

Section Details:

- Section ID: 5983505204379648
- Section Slug: design-improvements-of-a-web-crawler
- Status: Content pending generation

Use the section content generation API to create the content for this section.

5 Evaluation of Web Crawler's Design

Section content for "Evaluation of Web Crawler's Design" is being generated.

Section Details:

- Section ID: 6142975511363584
- Section Slug: evaluation-of-web-crawlers-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

CHAPTER 36

DESIGN WHATSAPP

1 System Design: WhatsApp . 170
2 Requirements of WhatsApps Design 170
3 High-level Design of WhatsApp 170
4 Detailed Design of WhatsApp 171
5 Evaluation of WhatsApps Design 171
6 Quiz on WhatsApps Design 171

Take a look at WhatsApp System Design with an emphasis on its API design, high security, and low latency of client-server messages.

1 System Design: WhatsApp

Section content for "System Design: WhatsApp" is being generated.

Section Details:

- Section ID: 6538438514049024
- Section Slug: system-design-whatsapp
- Status: Content pending generation

Use the section content generation API to create the content for this section.

2 Requirements of WhatsApps Design

Section content for "Requirements of WhatsApps Design" is being generated.

Section Details:

- Section ID: 5023065490063360
- Section Slug: requirements-of-whatsapps-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

3 High-level Design of WhatsApp

Section content for "High-level Design of WhatsApp" is being generated.

Section Details:

- Section ID: 6728651831246848
- Section Slug: high-level-design-of-whatsapp
- Status: Content pending generation

Use the section content generation API to create the content for this section.

4 Detailed Design of WhatsApp

Section content for "Detailed Design of WhatsApp" is being generated.

Section Details:

- Section ID: 6731604231192576
- Section Slug: detailed-design-of-whatsapp
- Status: Content pending generation

Use the section content generation API to create the content for this section.

5 Evaluation of WhatsApps Design

Section content for "Evaluation of WhatsApps Design" is being generated.

Section Details:

- Section ID: 4861549654573056
- Section Slug: evaluation-of-whatsapps-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

6 Quiz on WhatsApps Design

Section content for "Quiz on WhatsApps Design" is being generated.

Section Details:

- Section ID: 4877609447456768
- Section Slug: quiz-on-whatsapps-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

CHAPTER 37

DESIGN TYPEAHEAD SUGGESTION

1 System Design: The Typeahead Suggestion System . . . 174
2 Requirements of the Typeahead Suggestion Systems Design 174
3 High-level Design of the Typeahead Suggestion System . 174
4 Data Structure for Storing Prefixes 175
5 Detailed Design of the Typeahead Suggestion System . 175
6 Evaluation of the Typeahead Suggestion Systems Design . 175
7 Quiz on the Typeahead Suggestion Systems Design . . . 176

Discover the typeahead suggestion System Design, covering aspects like efficient data structures and assemblers for updating suggestions.

1 System Design: The Typeahead Suggestion System

Section content for "System Design: The Typeahead Suggestion System" is being generated.

Section Details:

- Section ID: 4981044919140352
- Section Slug: system-design-the-typeahead-suggestion-system
- Status: Content pending generation

Use the section content generation API to create the content for this section.

2 Requirements of the Typeahead Suggestion Systems Design

Section content for "Requirements of the Typeahead Suggestion Systems Design" is being generated.

Section Details:

- Section ID: 6369391511339008
- Section Slug: requirements-of-the-typeahead-suggestion-systems-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

3 High-level Design of the Typeahead Suggestion System

Section content for "High-level Design of the Typeahead Suggestion System" is being generated.

Section Details:

- Section ID: 6376843531255808
- Section Slug: high-level-design-of-the-typeahead-suggestion-system

- Status: Content pending generation

Use the section content generation API to create the content for this section.

4 Data Structure for Storing Prefixes

Section content for "Data Structure for Storing Prefixes" is being generated.

Section Details:

- Section ID: 6110217372172288
- Section Slug: data-structure-for-storing-prefixes
- Status: Content pending generation

Use the section content generation API to create the content for this section.

5 Detailed Design of the Typeahead Suggestion System

Section content for "Detailed Design of the Typeahead Suggestion System" is being generated.

Section Details:

- Section ID: 5464491722801152
- Section Slug: detailed-design-of-the-typeahead-suggestion-system
- Status: Content pending generation

Use the section content generation API to create the content for this section.

6 Evaluation of the Typeahead Suggestion Systems Design

Section content for "Evaluation of the Typeahead Suggestion Systems Design" is being generated.

Section Details:

- Section ID: 6626894497185792
- Section Slug: evaluation-of-the-typeahead-suggestion-systems-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

7 Quiz on the Typeahead Suggestion Systems Design

Section content for "Quiz on the Typeahead Suggestion Systems Design" is being generated.

Section Details:

- Section ID: 5201949791617024
- Section Slug: quiz-on-the-typeahead-suggestion-systems-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

CHAPTER 38

DESIGN A COLLABORATIVE DOCUMENT EDITING SERVICE / GOOGLE DOCS

1 System Design: Google Docs 178
2 Requirements of Google Docs Design 178
3 Design of Google Docs . . . 178
4 Concurrency in Collaborative Editing 179
5 Evaluation of Google Docs Design 179

Understand the System Design of Google Docs, using different techniques to address storage, collaborative editing, and concurrency issues.

1 System Design: Google Docs

Section content for "System Design: Google Docs" is being generated.

Section Details:

- Section ID: 5540995072000000
- Section Slug: system-design-google-docs
- Status: Content pending generation

Use the section content generation API to create the content for this section.

2 Requirements of Google Docs Design

Section content for "Requirements of Google Docs Design" is being generated.

Section Details:

- Section ID: 5975395656007680
- Section Slug: requirements-of-google-docs-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

3 Design of Google Docs

Section content for "Design of Google Docs" is being generated.

Section Details:

- Section ID: 5921693154934784
- Section Slug: design-of-google-docs
- Status: Content pending generation

Use the section content generation API to create the content for this section.

4 Concurrency in Collaborative Editing

Section content for "Concurrency in Collaborative Editing" is being generated.

Section Details:

- Section ID: 6200295243120640
- Section Slug: concurrency-in-collaborative-editing
- Status: Content pending generation

Use the section content generation API to create the content for this section.

5 Evaluation of Google Docs Design

Section content for "Evaluation of Google Docs Design" is being generated.

Section Details:

- Section ID: 5489039691481088
- Section Slug: evaluation-of-google-docs-design
- Status: Content pending generation

Use the section content generation API to create the content for this section.

CHAPTER 39

SPECTACULAR FAILURES

1 Introduction to Distributed System Failures 182
2 Facebook, WhatsApp, Instagram, Oculus Outage 182
3 AWS Kinesis Outage Affecting Many Organizations . . . 182
4 AWS Wide Spread Outage . 183

Learn how to avoid failures and outages by discussing case studies of real-world, large-scale distributed systems like Facebook, AWS, etc.

1 Introduction to Distributed System Failures

Section content for "Introduction to Distributed System Failures" is being generated.

Section Details:

- Section ID: 6503055815606272
- Section Slug: introduction-to-distributed-system-failures
- Status: Content pending generation

Use the section content generation API to create the content for this section.

2 Facebook, WhatsApp, Instagram, Oculus Outage

Section content for "Facebook, WhatsApp, Instagram, Oculus Outage" is being generated.

Section Details:

- Section ID: 5214324309622784
- Section Slug: facebook-whatsapp-instagram-oculus-outage
- Status: Content pending generation

Use the section content generation API to create the content for this section.

3 AWS Kinesis Outage Affecting Many Organizations

Section content for "AWS Kinesis Outage Affecting Many Organizations" is being generated.

Section Details:

- Section ID: 5243362298298368
- Section Slug: aws-kinesis-outage-affecting-many-organizations
- Status: Content pending generation

Use the section content generation API to create the content for this section.

4 AWS Wide Spread Outage

Section content for "AWS Wide Spread Outage" is being generated.

Section Details:

- Section ID: 6672582694141952
- Section Slug: aws-wide-spread-outage
- Status: Content pending generation

Use the section content generation API to create the content for this section.

CHAPTER 40

CONCLUDING REMARKS

1 Conclusions 186
2 Resources to Prepare for a System Design Interview . . . 186
1 No Sections Available 187

Reflect on key takeaways, highlight unique aspects of each Design Problem, and get pointers on the next steps to master System Design.

1 Conclusions

Section content for "Conclusions" is being generated.

Section Details:

- Section ID: 5067216508157952
- Section Slug: conclusions
- Status: Content pending generation

Use the section content generation API to create the content for this section.

2 Resources to Prepare for a System Design Interview

Section content for "Resources to Prepare for a System Design Interview" is being generated.

Section Details:

- Section ID: 5546916426809344
- Section Slug: resources-to-prepare-for-a-system-design-interview
- Status: Content pending generation

Use the section content generation API to create the content for this section.

CHAPTER 41

RECOVERED LESSONS

1 No Sections Available

This chapter contains no sections.